

Lab 6 — Containerize and Serve a Detector

Track B (BMD-45 vehicle detection) · Week 6 · DS5619 Machine Learning Systems Operations

Setup

```
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
python generate_for_student.py --student-id <your roll number or institute email>
```

This overwrites `data/fixtures/` with your own synthetic CCTV images — same file names, same two camera profiles, always at least one detectable "vehicle" per image, generated deterministically from your student ID. Two students never get the same pixels.

Record your `--student-id` value in `NOTES.md`. The grader re-runs `generate_for_student.py` with the ID you recorded and diffs the result against what you committed.

Files

- `src/app.py` — implement the `# TODO` functions/route.
- `src/mock_detector.py` — dependency-light stand-in detector (provided, don't edit — see its docstring for why it exists and how to swap it out).
- `Dockerfile` — fill in the `???` blanks.
- `data/fixtures/` — your personalized synthetic "CCTV" images, generated above, for local/container testing.

Why a mock detector

BMD-45 (the CCTV vehicle-detection dataset this track is built around, paper: <https://huggingface.co/datasets/iisc-aim/BMD-45>) ships real checkpoints (YOLOv12, RT-DETRv2, D-FINE, ...), but pulling one in plus torch plus a GPU is a heavy, slow dependency chain for a 110-minute session about *packaging*, not model accuracy. `src/mock_detector.py` (complete, don't edit — read its docstring) is a trivial stand-in behind the same interface a real checkpoint would use. `data/fixtures/` has small synthetic "CCTV" images it can actually detect something on.

Your task

Part 1 — `src/app.py` (three functions/routes marked `# TODO`, ~40 min)

- `load_image_from_upload(file_storage)` — turn an uploaded file into a PIL Image.
- `run_detection(image)` — run the mock detector, return a JSON-serializable result.
- the `/detect` route — wire the above together, handle the missing-file case with a 400.

Run it locally first to sanity check before touching Docker:

```
python src/app.py
# in another terminal:
curl http://localhost:8080/health
curl -F "image=@data/fixtures/camera_A_daylight/000.jpg"
http://localhost:8080/detect
```

Part 2 — Dockerfile (five ??? blanks, ~25 min)

Fill in the base image, the dependency install step, the source copy step, and the start command. Then:

```
docker build -t week6-detector .
docker run --rm -p 8080:8080 week6-detector
# in another terminal, same curl commands as above, against the container
this time
```

Part 3 — DOCKER_VERIFICATION.md (~15 min)

Paste your actual build/run/curl commands and their output. This is how we confirm the container works without needing Docker-in-Docker in an automated grader.

Self-check

```
pytest tests/ -q
```

Tests the Flask app directly (no Docker needed for this part) — a self-check, not the grader.

Deliverables (what you commit)

- `src/app.py`, completed.
- `Dockerfile`, completed.
- `DOCKER_VERIFICATION.md`, filled in with real output (not placeholder text).
- A short `NOTES.md`: the `--student-id` value you used (required — see above), what image size did `docker images` report for `week6-detector`, and what's the single biggest thing you'd change about this Dockerfile if `src/mock_detector.py` were swapped for a real torch-based checkpoint (think about what that does to build time and image size)?

Grading checklist

- ☐ `data/fixtures/` matches what `generate_for_student.py --student-id <NOTES.md value>` actually produces.
- ☐ `/health` and `/detect` work correctly against the Flask test client (checked by `tests/test_smoke.py`, which is not the full grader).
- ☐ `/detect` returns a 400 with a clear error when no image is uploaded.
- ☐ `Dockerfile` builds successfully and the container serves both endpoints correctly (verified via `DOCKER_VERIFICATION.md`'s pasted output, cross-checked by the grader rebuilding it).

- ☐ **NOTES.md** shows you actually looked at the built image size, not a guess.
- ☐ Meaningful commit history and a working README.

Submission

```
git add -A
git commit -m "Week 6: containerize + serve detector"
git tag week06-submit
git push origin main --tags
```